Critical Literature Review: Dynamic Player Ratings, Matchmaking, and Mobile Amateur Tennis Tournament Management

Timur Bizinskiy, Arlan Tulegenov

I. Introduction

Digital technologies are increasingly used in sport to support performance evaluation, competition management, communication, and data-driven decision-making. Sports recommender systems can support athletes and organizers through recommendations related to training, teams, opponents, tactics, and other sport-specific decisions [1]. Mobile applications also provide a practical interface for registration, event management, communication, and access to performance information. Reviews of sports applications show that such systems are used for performance monitoring, skill development, athlete self-management, and other sport-related tasks, while also revealing limitations in validation, reporting, and long-term evaluation [2], [3].

The proposed project focuses on a mobile application for an amateur tennis community that supports both players and tournament organizers. Players should be able to register for amateur tournaments, view their rating and schedule, and receive information about upcoming matches. Organizers should be able to manage participants, form tournament draws or groups, seed players according to competitive level, manage match schedules and results, with more advanced court allocation and rescheduling considered as possible extensions. The central analytical component of the system is a dynamic player rating. Instead of relying only on a manually assigned level, the application is intended to update player strength from match results using rating approaches such as Elo and Glicko-2.

This creates several connected engineering and research problems. First, a rating system must represent player strength accurately enough to support fair seeding and competitive pairings, despite noisy or sparse amateur match histories. Second, amateur players may participate irregularly or improve rapidly, so a useful model should respond to uncertainty and changes in performance. Third, tournaments operate under practical constraints such as limited time, player availability, and avoiding invalid duplicate registrations or matches. In doubles, team formation introduces further constraints. Finally, all of these functions must be integrated into a usable mobile system for both organizers and players.

For this reason, the literature is organized into five themes: dynamic rating and skill estimation; validation of tennis ratings; opponent and doubles partner matching; tournament design and scheduling; and mobile sports platforms with decision support. Together, these areas provide the theoretical and technical foundation for a system that uses dynamic ratings not only to describe player skill, but also to support tournament organization.

II. Dynamic Rating and Skill Estimation

A central requirement of the proposed application is the ability to update player skill after each recorded match. Rating systems differ from simple ranking tables because they attempt to estimate underlying competitive strength rather than only accumulate tournament points. Glickman and Jones [4] review statistical models for head-to-head competition and explain how dynamic paired-comparison models can be simplified into practical rating systems such as Elo and Glicko. Their review shows that the main problem is not only predicting a single result, but continuously estimating player or team strength as it changes over time.

Elo provides one of the simplest approaches to this problem. A player's rating is updated according to the expected result of a match, the observed result, and a parameter controlling the size of the update. Kelley [5] examined Elo in tennis by comparing it with the Davis Cup ranking system through simulation. The systems were evaluated using criteria including correlation with competitive performance, responsiveness to improvement, and protection against temporary decline. Standard Elo responded relatively slowly when a competitor improved, whereas an alternative optimization of the K-factor produced performance comparable to the Davis Cup approach. This result is important for amateur tennis because it shows that Elo is interpretable and computationally simple, but its behavior depends strongly on parameter selection. A K-factor suitable for frequent professional competition may not necessarily be suitable for amateur players who play at very different frequencies.

Glicko-type models extend rating by representing uncertainty about the current estimate. Glickman [6] developed a dynamic paired-comparison model in which the variance of a competitor's underlying strength can itself change over time. The model was designed to handle situations in which ability may change gradually or abruptly, and it was demonstrated using game data from chess and American football. This work provides an important statistical foundation for the development of Glicko-2, where rating is accompanied by rating uncertainty and a volatility component. For amateur tennis, this is particularly relevant because players may remain inactive for long periods or may improve rapidly. In such

situations, it is useful for the system to distinguish between a stable rating supported by many recent matches and a rating whose current accuracy is uncertain.

The use of Glicko in tennis has also been tested empirically. Yue et al. [7] studied men's and women's Grand Slam singles matches from 2000 to 2019 and compared Glicko-derived variables with conventional rankings and several machine-learning models. Official rankings already contained substantial predictive information, but Glicko-based approaches achieved slightly higher predictive accuracy, and court-related information improved ATP prediction further. The study shows that a dynamic rating can provide useful predictive information without requiring an excessively complex model.

Taken together, the Elo and Glicko literature suggests an important design trade-off. Elo is transparent and easy to implement, but its responsiveness depends on a fixed update parameter. Glicko-2 is more complex, but it explicitly models uncertainty and changing performance. This difference provides a strong basis for comparing the two approaches in an amateur setting where match frequency is irregular and the quality of an initial manually assigned rating may vary.

III. Rating Validation, Ranking, and Tournament Seeding

A rating is useful for tournament management only if it represents competitive level with reasonable accuracy. Existing tennis research commonly evaluates rating quality by testing whether rating differences predict subsequent match outcomes. This approach is directly relevant to the proposed system because the rating will be used as a basis for seeding, grouping, and selecting competitive opponents.

Mayew and Mayew [8] compared Universal Tennis Rating (UTR) and World Tennis Number (WTN) using 1,532 matches played by 870 participants at the 2022 USTA Junior National Championships. They treated the ability to predict head-to-head match outcomes as evidence of rating quality. UTR predicted winners with 73.9% classification accuracy and WTN with 70.4%; both exceeded chance, but the difference between the two systems was not statistically significant. The authors explicitly connect rating quality to practical decisions such as tournament seeding and the creation of competitively balanced matches. A strength of this study is that the ratings were evaluated on a large real tournament dataset. A limitation is that the sample consisted of high-level junior players, so the results cannot automatically be generalized to recreational adults.

Kiely et al. [9] extended this line of research after an update to the WTN algorithm. Their study compared updated WTN and UTR ratings across elite junior and collegiate competitions. The results did not show consistent evidence that WTN outperformed UTR; UTR demonstrated stronger predictive accuracy in several junior settings and in some collegiate stages. This reinforces the conclusion that rating performance is population-dependent and that claims about a universally superior rating system should be treated cautiously.

Richey et al. [10] similarly compared UTR and WTN in junior and collegiate tennis. The systems performed similarly in the junior sample, whereas UTR showed stronger predictive performance in men's collegiate matches. They also found that the effect of rating differences could change depending on the competitive level of the players. These findings are important for an amateur system because the same rating parameters may not perform equally well for beginners, intermediate players, and stronger competitors.

The broader role of ranking and seeding is examined by Crespo et al. [11]. Their systematic review screened 363 records and included 14 studies related to tournament structure, player success, planning, ranking, and seeding. The review concludes that ranking and seeding systems have important predictive and evaluative roles in tennis competitions. However, amateur tournaments were excluded from the review's population criteria. This limitation is especially relevant to the proposed project because it shows that evidence from professional and junior tennis cannot simply be assumed to describe local amateur tournaments.

For the proposed application, these studies support a distinction between an *initial level* and a *dynamic rating*. A player may self-assess an initial NTRP-inspired level before sufficient match history exists, while Elo or Glicko-2 can maintain an internal continuous rating that changes after recorded results. The system can then map this dynamic estimate back to user-facing skill categories. This approach makes it possible to study whether the initial self-assessed level is consistent with later match evidence and whether an automated rating converges toward a more reliable estimate of competitive strength.

IV. Opponent Matching and Doubles Partner Formation

The diploma project's automatic matching function requires distinguishing two related decisions: choosing competitively suitable opponents and forming doubles teams from individual registrants. Research on head-to-head rating supports using estimated skill differences as one input to opponent selection, but match-outcome prediction is not equivalent to proving that a pairing is desirable to players. A close expected result does not by itself capture availability, preferred tournament format, or willingness to play together. The dynamic-rating and validation evidence therefore provides a basis for measurable balance criteria, rather than a complete matchmaking algorithm [4], [7], [8].

Sports recommender-system research helps frame matching as a decision with multiple criteria. Felfernig et al. [1] describe recommendation tasks involving teams and other sport-specific choices, while Mateos and Bellogín [12] discuss contextual factors that may affect recommendations. Roy and Dutta [13] identify cold-start and sparse-data problems that are directly relevant when a new player has no match history. These studies support considering both the estimated playing level and explicit registration constraints; they do not independently validate any particular amateur-tennis doubles pairing formula.

For the planned system, players may register for Doubles with an already agreed partner or enter as individuals eligible for automatic pairing. Confirmed premade teams should not

be broken up by the algorithm. For individual registrants, the pairing method can aim to avoid large skill imbalances while enforcing unique assignment and confirmation rules. The team will need to decide whether balancing means minimizing differences *within* a doubles team, balancing the estimated strength of opposing teams, or combining these objectives. A further methodological issue is that singles rating evidence does not directly establish how to calculate a doubles-team rating or attribute a doubles result to each partner; this must be treated as a separate assumption to be evaluated. Consequently, the review supports a modest first implementation with explainable constraints and measurable balance, rather than claiming that the literature already establishes a proven optimal pairing method.

V. Tournament Design, Seeding, and Scheduling

Rating is only one component of tournament organization. Even when participants have been evaluated correctly, organizers must decide how players are seeded or grouped and how matches are assigned to courts and time slots. Devriesere et al. [14] review tournament design from an operational-research perspective and divide the problem into five major components: format, seeding, draw, scheduling, and ranking. They evaluate these components in relation to three general objectives: efficacy, fairness, and attractiveness. This framework is useful for the proposed application because a tournament-management system should not optimize only one task. Accurate ratings contribute to fair seeding, while scheduling must also respect operational constraints.

Sports timetabling is computationally difficult because different competitions use different rules and constraints. Van Bulck et al. [15] analyzed more than 500 generated sports-timetabling instances and compared eight state-of-the-art algorithms. Their results showed that no single algorithm performs best for every problem type. They therefore developed an algorithm-selection system that predicts which scheduling method is most suitable based on the competition type and constraints. The study demonstrates that scheduling should be treated as a constraint-dependent optimization problem rather than as a fixed template. For an amateur tennis tournament, relevant constraints may include the number of courts, match duration, player availability, rest time, and the remaining duration of the event.

Li and Goossens [16] study the integration of grouping and scheduling rather than treating them as independent stages. Their multi-league grouping and scheduling problem balances two conflicting objectives: minimizing travel distance and minimizing violations of shared venue capacity. They formulate a bi-objective mixed-integer programming model and develop heuristic methods because the combined problem is computationally difficult. Although their application concerns sports leagues rather than tennis tournaments, the underlying principle is directly transferable: assigning competitors to groups and assigning their matches to limited facilities are interdependent decisions.

The operational importance of time and resources is also emphasized in the sports-logistics literature. Pott et al. [17] systematically review logistics in sport and propose a framework covering the movement and availability of people, equipment, facilities, and information. Their review emphasizes that sporting events depend on delivering resources at the correct time and place, while also noting that sport logistics remains comparatively under-researched. In a one-day amateur tennis tournament, this issue becomes especially visible because unused court time or delayed match assignment can reduce the number of matches that can be completed.

Context-aware recommendation provides an additional perspective on dynamic tournament decisions. Mateos and Bellogín [12] identify temporal, spatial, and social context as major categories used by context-aware recommender systems, while Venkatachalam and Ray [18] show how contextual information can be applied in sports and fitness recommendation. Afzal et al. [19] provide a technical example in which multiple contextual variables and multiple criteria are processed within one deep-learning recommender model. Their datasets are not related to tennis, so the results cannot be transferred directly; however, the study demonstrates that recommendations can account for several simultaneous criteria rather than a single score.

For the proposed application, this literature supports rule-based generation and display of a feasible tournament match schedule for one selected competition system as a realistic initial objective. Advanced allocation of courts or dynamic rescheduling can be investigated later; it is not necessary to assume that the first working release will solve a general optimization problem. If introduced, a recommendation for the next available match would need to respect tournament rules, eligibility, rest constraints, and court availability. The reviewed literature does not by itself demonstrate that a real-time scheduling optimizer is needed in the target amateur community; user interviews should test this assumption.

VI. Mobile Sports Platforms and Decision Support

The final system must present rating and tournament-management functions through a usable mobile platform. Purwanto et al. [20] developed SIMPORA, an Android-based framework for community sports data management and analytics. The system integrates user, event, communication, and sports-data functions and demonstrates how mobile technology can centralize information that would otherwise be managed through separate manual processes. This is relevant to the proposed tennis application because tournament registration, participant records, match results, ratings, and schedule changes can all be maintained in one environment.

Sports recommender research provides a broader view of the types of decisions such a platform may support. Felfernig et al. [1] review recommender systems used for training, teams, tactics, nutrition, and other sport-related decisions. Their work shows that sports recommendation is not limited to consumer-style item recommendation and can instead support domain-specific choices. Roy and Dutta [13] review general recommender-system approaches and identify cold start, sparsity, and scalability as persistent problems. Cold start is especially important for a tournament platform because

a new player may join with no recorded match history. A self-assessed initial level can therefore act as a temporary prior estimate until enough results are available for a dynamic rating.

Mobile-sport research also shows that technical correctness alone is not sufficient. Angosto et al. [21] review factors affecting intention to use fitness applications and identify perceived usefulness, ease of use, enjoyment, social influence, and other behavioral variables as recurring influences on adoption. Barisch-Fritz et al. [3] examine app development in a sports-science setting and report inconsistent use and documentation of software-engineering practices, including requirements, testing, security, and technology decisions. These findings suggest that the application should be designed not only as an algorithmic prototype, but also as a reliable tool for organizers and players.

Liu et al. [2] reviewed 111 studies of mobile applications in sports training and found that validation and reliability studies were much more common than controlled evaluations of whether apps improve real-world outcomes. This distinction is important for the proposed project. It is possible to demonstrate that a rating algorithm predicts match outcomes, yet still fail to show that the complete application improves tournament operation. A full evaluation should therefore distinguish algorithmic performance from system-level usefulness, such as whether organizers can create fair draws more consistently, reduce scheduling conflicts, and respond more quickly to available courts.

VII. Research Gap

The reviewed literature provides strong foundations for each major component of the proposed project, but these components are usually investigated separately. Dynamic rating research explains how competitive strength can be updated over time and how uncertainty can be represented [4], [6]. Tennis-specific studies evaluate Glicko, Elo, UTR, and WTN through match prediction and competitive performance [5], [7]–[10]. Tournament research examines ranking, seeding, format, scheduling, and operational constraints [11], [14]–[17]. Mobile and recommender-system studies provide methods for integrating contextual information, user management, decision support, and mobile interaction [1], [12], [13], [18]–[20].

However, several gaps remain when these findings are applied to amateur tennis tournaments. First, the reviewed tennis-rating studies evaluate professional, junior, collegiate, or established global rating systems, while evidence on dynamic Elo or Glicko-2 ratings for local amateur tournament participants is limited. Second, the literature does not establish how accurately a player's self-assessed starting level reflects a player's later competitive performance. Third, research on rating systems and research on tournament scheduling are largely separate, even though the quality of seeding and pairing decisions depends on the quality of the underlying player evaluation. Finally, the literature considered here gives limited direct evidence on forming amateur doubles partnerships while respecting player consent, rating uncertainty, and tournament rules. More advanced scheduling problems may emerge in future user research, but have not yet been established as essential to the first release.

These gaps suggest that the primary research focus should be the validity of the player-rating process rather than the mobile interface itself. A suitable central question is whether Elo and Glicko-2 can provide more accurate dynamic estimates of amateur tennis players' competitive level than players' initial self-assessed levels. Match outcomes can provide evidence for this comparison, while Glicko-2 additionally allows the system to represent uncertainty for players with limited or irregular match histories. Tournament seeding and scheduling then become the practical application context in which these ratings are used.

The proposed application can therefore integrate two related layers. The first layer maintains a dynamic player rating and evaluates its agreement with subsequent competitive results. The second layer uses this information to support tournament registration, seeding, doubles partner formation, opponent pairing, and match scheduling within an agreed first-release scope. This creates a clearer separation between the scientific question—how accurately player strength can be estimated—and the software objective—how that estimate can be used to make amateur tournament organization more efficient and consistent.

VIII. Implications for Engineering Design and Evaluation

The reviewed work motivates a testable separation between the rating algorithm, the matching policy, and the tournament workflow. Rating quality can be assessed through held-out chronological match predictions, using measures such as winner prediction accuracy and probabilistic error where suitable. Elo and Glicko-2 should be compared under comparable initial conditions and with transparent parameter choices. An initial self-assessment can serve as a provisional baseline, but only after the team defines a defensible mapping from ordinal displayed categories to the scale used by the algorithms. These are proposed evaluation procedures, not results reported for the future application.

The matching component should be evaluated independently of rating prediction. Possible criteria include the fraction of eligible individual Doubles registrants successfully paired, invalid or duplicate assignments, and the distribution of predicted team-strength differences. Player and organizer interviews should establish which indicators are meaningful before they become acceptance thresholds. Functional and integration tests must additionally verify registration, partner confirmation, match-result confirmation, and prevention of repeated rating updates for the same result. These choices translate the research into verifiable engineering requirements without assuming that predictive accuracy alone guarantees usability or tournament fairness.

The evidence base has limitations. Several tennis studies focus on elite juniors, collegiate players, or professional competition; their numerical outcomes should not be generalized to local recreational tennis. Research on sports timetabling and context-aware recommendation is technically relevant

but not direct empirical evidence for the target application. Studies of sports mobile apps also cannot establish adoption or satisfaction in this particular community. The proposed system should therefore use these publications to motivate design alternatives and conduct its own small-scale user validation and prototype testing. The initial release is not intended to establish an officially recognized NTRP score.

IX. Conclusion

The literature shows that dynamic player rating and tournament management are closely related but are usually studied as separate problems. Elo provides a simple and interpretable mechanism for updating competitive strength, while Glicko-2 extends dynamic rating by explicitly modelling uncertainty and changing variability. Tennis-specific studies demonstrate that rating quality can be evaluated through the prediction of match outcomes, but also show that performance depends on the population and competitive context.

Research on tournament design further demonstrates that ranking, seeding, draw formation, and scheduling influence fairness and operational efficiency. Sports-timetabling studies show that real competitions involve multiple constraints and that no single scheduling approach is optimal for every problem type. Reviews of sports logistics and mobile sports systems additionally show the importance of resource coordination, software quality, and usable decision support.

These findings support a revised project in which dynamic rating is the central analytical component of a mobile amateur-tennis tournament platform. A player's self-assessed NTRP-inspired level can provide an initial estimate for a new participant, while Elo and Glicko-2 can update an internal rating as match evidence accumulates. The resulting rating can then inform seeding, opponent matching, and doubles partner formation, while tournament scheduling is handled within the constraints of the selected format. The main research opportunity is therefore to evaluate whether automated dynamic ratings provide a more reliable representation of amateur player strength than self-assessed initial levels and to determine how this information can support more consistent tournament management.

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